

Midterm 2

● Graded

Student

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Total Points

19.5 / 50.5 pts

Question 1

Problem 1

3 / 6 pts

- 0 pts Correct

- 1 pt (a) $P(Y = 1|X \leq 1) = \frac{P(Y=1, X \leq 1)}{P(X \leq 1)} = \frac{P(X=0, Y=1) + P(X=1, Y=1)}{P(X \leq 1)}$: incorrect formulation

- 0.5 pts (a) $P(X = 0, Y = 1) + P(X = 1, Y = 1) = 0.25$: incorrect calculation

- 0.5 pts (a) $P(X \leq 1) = 0.8$: incorrect calculation

- 0.5 pts (a) $P(Y = 1|X \leq 1) = \frac{P(Y=1, X \leq 1)}{P(X \leq 1)} = \frac{P(X=0, Y=1) + P(X=1, Y=1)}{P(X \leq 1)} = 0.25 / 0.8 = 5/16$: incorrect final answer

✓ - 1 pt (b) $Var(X|Y = 1) = E[X^2|Y = 1] - E[X|Y = 1]^2$ or recognizing that $X|Y = 1 \sim Bern(p = P(X = 1|Y = 1))$ and $p = 2/5$: incorrect setup

✓ - 1 pt (b) $E[X|Y = 1] = 0P(X = 0|Y = 1) + 1P(X = 1|Y = 1) + 2P(X = 2|Y = 1) = P(X = 1|Y = 1) = 0.1/0.25 = 2/5$ and $E[X^2|Y = 1] = 0^2P(X = 0|Y = 1) + 1^2P(X = 1|Y = 1) + 2^2P(X = 2|Y = 1) = P(X = 1|Y = 1) = 0.1/0.25 = 2/5$: incorrect calculation

- 0.5 pts (b) get $Var(X|Y = 1) = p(1 - p) = 2/5(1 - 2/5) = 6/25$ through bernoulli distribution or $Var(X|Y = 1) = E[X^2|Y = 1] - E[X|Y = 1]^2 = 2/5 - (2/5)^2 = 6/25$: incorrect final answer

✓ - 1 pt (c) recognizing that $Y|X \leq 1 \sim Bern(p = P(Y = 1|X \leq 1))$ and $X|X \leq 1 \sim Bern(p = P(X = 1|X \leq 1))$ or calculate $E[Y|X \leq 1] = 5/16$ and $E[X|X \leq 1] = 1/2$: incorrect setup/calculation

- 1 pt (c) $E[16Y - 4X|X \leq 1] = 16E[Y|X \leq 1] - 4E[X|X \leq 1] = 3$: did not use linearity of expectation

- 0.5 pts Algebraic error

- 6 pts Wrong process

- 2 pts (b) incorrect setup, should have been $E[Y^2|X=0] - E[Y|X=0]^2$

- 2 pts (c) incorrect setup

- 1 pt (b) correct setup, wrong execution

- 2 pts (c) used linearity of expected value incorrectly, did not solve the problem

Question 2

Problem 2

5.5 / 8 pts

- 0 pts Correct
- 1 pt part a: incomplete
- 1 pt part a: correct set up but error in integrating
- 1.5 pts part a: incorrect setup
- 2.5 pts part a: incorrect set up and did not properly attempt the problem
- 3 pts part a: missing
- 0.5 pts part b: minor error in finding pdf of Y
- 0.5 pts part b: minor error in finding the variance of Y
- 1 pt part b: $Var(4Y + 3) = 16Var(Y)$
- 1 pt part b: incomplete

✓ – 1.5 pts part b: major error in finding variance

– 3 pts part b: missing

✓ – 1 pt part c: error in integration set up/integration

- 1 pt part c: incomplete
- 2 pts part c: missing
- 8 pts no work submitted
- 0.5 pts part c: most of calculations done correct, one error
- 1 pt part c: incorrect

Question 3

Problem 3

2 / 8 pts

- 1 pt $X = X_1 + X_2 + \cdots + X_{20}$

✓ - 1 pt $X_i \sim \text{Bern}(p = \frac{1}{20})$

✓ - 1 pt $\text{Var}(X_i) = \frac{19}{400}$, or $E(X_i^2) = \frac{1}{20}$

- 1 pt $\text{Var}(X) = \text{Cov}(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i)$, or $\text{Var}(X) = E(X^2) - E(X)^2$

✓ - 1 pt $\text{Cov}(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i) = \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$, or $E(X^2) = \sum_{i=1}^n E(X_i^2) + \sum_{i \neq j} E(X_i X_j)$

✓ - 1 pt $\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$

✓ - 1 pt $\sum_{i=1}^n \text{Var}(X_i) = \frac{19}{20}$, or $\sum_{i=1}^n E(X_i^2) = 1$

✓ - 1 pt $\sum_{i \neq j} \text{Cov}(X_i, X_j) = \frac{1}{20}$, or $\sum_{i \neq j} E(X_i X_j) = 1$

Click here to replace this description.

- 0 pts Correct

- 8 pts Incorrect

- 1 pt final answer

- 1 pt Calculation mistake

Question 4

4.5 / 9 pts

Problem 4

- 0 pts Correct

- 1 pt part a: error in the covariance term/did not include covariance in final simplification

- 1.5 pts part a: did not find variance of X and Y correctly

✓ - 2 pts part a: did not simplify variance correctly

- 3 pts part a: missing

- 0.5 pts part b: did not use 68-95-99 rule correctly to simplify to decimal

- 1 pt part b: work not shown

- 1 pt part b: z score not calculated correctly

✓ - 1.5 pts part b: did not use Z scores to simplify to a number (express the probability)

- 3 pts part b: part missing

- 1 pt part b: found $P(|Y| < 2)$ not $P(Y < 2)$

- 0.5 pts part c: did not use the 68-95-99 rule correctly to simplify to a number

- 0.5 pts part c; calculation error

- 1 pt part c: incomplete

✓ - 1.5 pts part c: major error in problem

- 2 pts part c: started part but did not get far

- 3 pts part c: missing

💬 + 0.5 pts did not take the absolute value probability correctly but almost got it

Question 5

Problem 5

1 / 6 pts

- 1 pt $\lambda = 2$ ✓ - 1 pt $P(T > 1) = \int_1^{\infty} 2e^{-2t} dt$ ✓ - 1 pt $\int_1^{\infty} 2e^{-2t} dt = e^{-2}$ ✓ - 1 pt $X \sim \text{Bin}(n = 6, p = e^{-2})$ ✓ - 1 pt $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$ ✓ - 1 pt Answer: $(1 - e^{-2})^6 + 6e^{-2}(1 - e^{-2})^5 + 15e^{-4}(1 - e^{-2})^4$

Click here to replace this description.

- 0 pts Correct

- 6 pts Incorrect

Question 6

Problem 6

0 / 6 pts

- 0 pts Correct

✓ - 2 pts Solution1: probability of defective = $P(D) = P(D|A)P(A) + P(D|B)P(B) = 0.2 * 0.3 + 0.05 * 0.7 = 0.095$: incorrect defective probability

✓ - 2 pts Solution1: $P(A|3 \text{ defective in } 10) = \frac{P(A \text{ and } 3 \text{ defective in } 10)}{P(3 \text{ defective in } 10)}$: incorrect setup

✓ - 1 pt Solution1: $P(A \text{ and } 3 \text{ defective in } 10) = P(3 \text{ defective in } 10|A)P(A) = \binom{10}{3}(0.2)^3(0.8)^7 0.3$: incorrect probability

✓ - 1 pt Solution1: $P(3 \text{ defective in } 10) = \binom{10}{3}(0.095)^3(0.905)^7$: incorrect probability

- 0.5 pts algebraic error

- 4 pts wrong process. here you are assuming all 10 bulbs are from branch A, but that is uncertain. Giving you 2 pts since you showed a correct use of the bayes theorem.

- 1 pt unrelated 4 in binomial distribution

- 4 pts Solution2: incorrect $P(3 \text{ defective in } 10|A)$ and $P(3 \text{ defective in } 10|B)$

- 1 pt Unrelated 20

- 1 pt Solution2: incorrect $P(A)P(3 \text{ defective in } 10|A)$

- 1 pt Solution2: incorrect $P(B)P(3 \text{ defect in } 10|B)$

- 6 pts cannot follow, no logic

- 6 pts no solution

Question 7

Problem 7

1.5 / 3 pts

✓ - 0.5 pts (a) distribution and its parameters

✓ - 0.5 pts (a) setup of probability

- 0.5 pts (b) distribution and its parameters

✓ - 0.5 pts (b) setup of probability

- 0.5 pts (c) distribution and its parameters

- 0.5 pts (c) setup of probability

Click here to replace this description.

- 0 pts Correct

- 3 pts Incorrect

Question 8

Problem 8

1 / 3 pts

– 0 pts Correct

– 1 pt part a incorrect

✓ – 1 pt part b incorrect

✓ – 1 pt part c incorrect

– 1.5 pts did not do probability set up for all 3 but all three distributions correct

– 1.5 pts consistent parameter error carried throughout

– 3 pts missing

Question 9

Problem 9

1 / 1.5 pts

– 0 pts Correct

– 0.5 pts (a) False

✓ – 0.5 pts (b) False

– 0.5 pts (c) Negative

* **Problem 1.** [Total 6 points] For each part simplify your final answer as much as possible.

	$X=0$	$X=1$	$X=2$
$Y=1$	0.15	0.1	0
$Y=0$	0.25	0.3	0.2

$$0.15 + 0.25 = 0.40$$

$$+ \quad = 0.80$$

$$0.1 + 0.3 = 0.40$$

$$E[X^2] - (E[X])^2$$

(a) Find $P(Y=1|X \leq 1)$.

(b) Find $\text{Var}(X|Y=1)$. $p(1-p)$ $0.25(0.75)$

(c) Find $E[16Y - 4X|X \leq 1]$ $\rightarrow 0,1 \rightarrow 16(0.80) - 4(0.80)$ $\mu = p$

Bernoulli
r.v

$$a) \quad P(Y=1|X \leq 1) = \frac{P(Y=1, X \leq 1)}{P(X \leq 1)} = \frac{0.25}{0.80} = \frac{25}{80} = \frac{5}{16}$$

$\hookrightarrow X=0,1$

$$b) \quad \text{Var}(X|Y=1) = \frac{\text{Var}(X, Y=1)}{\text{Var}(Y=1)} = \frac{(0.25)(0.75)}{\frac{25}{100} \cdot \frac{75}{100}}$$

$X = 0 \cdot 0.15 + X + 0.1 + X + 0$

$$c) \quad E[16Y - 4X|X \leq 1] = \frac{E[16Y - 4X, X \leq 1]}{E[X \leq 1]} = \frac{16(0.80) - 4(0.80)}{16 \cdot \frac{4}{5} - 4 \cdot \frac{4}{5}}$$

$$\frac{64}{5} - \frac{16}{5} = \frac{48}{5}$$

Problem 2. [Total 8 points] Show all your work.

* Let the joint probability density function for X and Y be given as

$$f(x, y) = \begin{cases} Cx(1-x)e^{-2y} & 0 < x < 1, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the constant C to make the above a valid pdf.

(b) Find $\text{Var}(4Y+3)$. $\rightarrow 16\text{Var}(Y)$

$$E[Y^2] - (E[Y])^2$$

(c) Find $\text{Cov}(X, Y)$. $\rightarrow E[XY] - E[X]E[Y]$

$$a) C \int_0^1 \int_0^\infty x(1-x)e^{-2y} dy dx = C \int_0^1 \int_0^\infty x - x^2 e^{-2y} dy dx$$

$$= C \int_0^\infty e^{-2y} dy = -\frac{e^{-2y}}{2} \Big|_0^\infty = \frac{1}{2} \quad \text{inner}$$

$$\int_0^1 x - x^2 dx = \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \quad \text{outer}$$

$$C \cdot \frac{1}{2} \cdot \frac{1}{6} = 1 \rightarrow C = 12$$

$$b) 12 \int_0^\infty x - x^2 e^{-2y} dy dx = \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^\infty = 1 \quad ?$$

$$16 \text{Var}(Y) = 16 \cdot 12$$

$$c) \text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$12 \int_0^1 \int_0^\infty xy (x - x^2 e^{-2y}) dy dx = 12 \int_0^1 \int_0^\infty x^2 y - x^3 y e^{-2y} dy dx$$

$$= \frac{6}{4}$$

Problem 3. [8 points] There are 20 students in a classroom. Each student hands in their homework in a folder with their name on it. The teacher shuffles the folders and hands them back randomly, one to each student.

Linearity of exp.

Let X be number of students who get back their own homework folder. Calculate $\text{Var}(X)$.
SHOW ALL YOUR WORK.

$$X_1 + X_2 + \dots + X_n$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$2 \sum_{i \neq j} E[X_i] E[X_j]$$

$$\sum_{i=1}^{n=20} E[X_i] = \frac{1}{20}$$

$$\sum_{j=1}^{n=19} E[X_j] = \frac{1}{19}$$

$$= 20 \cdot 19$$

$$\text{Var}(X) = \frac{1}{20} \cdot \frac{1}{19} \cdot \frac{1}{19}$$

$$\frac{1}{20} \left(\frac{1}{19} \right)^2$$

Problem 4. The followings are for Part(a) and (b) (μ, σ^2)

$$E[X] = \mu_X = 1, \quad \text{Var}(X) = \sigma_X^2 = 4, \quad X \sim N(\mu_X, \sigma_X^2)$$

$$E[Y] = \mu_Y = -2, \quad \text{Var}(Y) = \sigma_Y^2 = 9$$

Evaluate the followings. SHOW ALL YOUR WORK.

$$z = \frac{X - \mu}{\sigma}$$

$$(a) \text{Var}\left(3X + \frac{Y}{3}\right) \rightarrow 9\text{Var}(X) + \text{Var}\left(\frac{Y}{3}\right)$$

* (b) Find the value of Cdf of X at X = 5, meaning $F_X(5)$ (c) Find $P(2 < |X - 1| < 4)$

$$a) \text{Var}\left(3X + \frac{Y}{3}\right) \rightarrow (9 \cdot 4) + \frac{9}{3} \rightarrow 36 + 3 = 39$$

$$b) F_X(5) = F(5) - F(5^-) = 3$$

$$c) P(-2 < X - 1 < 4) = P(3 < X < 5) = P\left(\frac{3-1}{2} < Z < \frac{5-1}{2}\right)$$

$$= P(1 < Z < 2) = \Phi(2) - \Phi(1) = \frac{0.95 - 0.68}{2} = 0.27$$

these are
not Laplace

Problem 5. [6 points] Let T denote the time it takes for a computer to shut down. Suppose T follows an Exponential distribution with mean 0.5 minutes. A computer lab has 6 independent computers that must all be shut down at the end of the day.

$$uv - \int v du$$

What is the probability that at most 2 computers take longer than 1 minute to shut down on a given day?

$$\mu = 0.5$$

$$0.5\lambda = 1$$

$$\mu = \frac{1}{\lambda} = \frac{1}{0.5} = \frac{1}{\frac{1}{2}} = 2$$

$$\frac{1}{\lambda} = 0.5$$

$$\frac{1}{\frac{1}{2}} = 2$$

$$P(X \leq 2) = \int_0^2 \lambda e^{-\lambda x} dx = \int_0^2 2e^{-2x} dx$$

low: 0

up: -4

$$U = -2x \quad dv = -2dx$$

$$V = 2e^{-2x} \quad \frac{dv}{dx} = -2$$

$$dv = \frac{e^{-2x}}{-2}$$

$$-2e^{-2x} \cdot 2x - \int -4e^{-2x} dx = \frac{1}{2}$$

L'Hopital

$$2e^{-2(6)} = 2e^{-12}$$

	D	∫
+	-2x	2e ^{-2x}
-	-2	e ^{-2x}
+	0	e ^{-2x}
		4

Problem 6.[6 points] A factory produces light bulbs. This factory has two branches Branch A and Branch B. The probability of a bulb being defective if it is made in Branch A is 20%, while the probability of a bulb being defective if it is made in Branch B is 5%. Assume 30% of the products are made in Branch A.

A random sample of 10 bulbs was inspected, and 3 of them were found to be defective. What is the probability that these bulbs had been made in Branch A.

$$P(D|A) = 0.20$$

$$P(D|B) = 0.05$$

$$P(A) = P(D|A)P(A) \\ = 0.20 \cdot 0.30 = 0.06$$

$$\binom{10}{3} (0.30)^3 (0.70)^{10-3} \\ = \binom{10}{3} (0.30)^3 (0.70)^7$$

Problem 7. Sarah works as a sales representative selling house insurance over the phone. Only 10% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.

For each part only state the name of distribution and its parameters and set up your probability no need to evaluate anything, for example write: " $X \sim \text{Bin}(n = 10, p = 0.2)$ " and my answer to this question is $P(X \geq 2)$ "

$$\binom{9-1}{3-1}$$

- a. [1 point] What is the probability that Sarah makes her second sale on the 9th call?
 b. [1 point] What is the probability that Sarah makes her first sale on the third call?
 c. [1 point] What is the probability that out of 20 call Sarah makes at least 3 sales?

a) $X \sim \text{NBin}(r=3, p=0.10), P(X=2) \quad \binom{9-1}{3-1} (0.10)^3 (0.90)^6$
 b) $X \sim \text{Geom}(p=0.10), P(X=1) \quad (0.10)(0.90)^2$
 c) $X \sim \text{Bin}(n=20, p=0.10), 1 - P(X \leq 3) \quad 1 - \sum_{x=0}^3 \binom{20}{x} (0.10)^x (0.90)^{20-x}$

Problem 8. A coffee shop receives, on average, 10 customers per hour.

For each part only state the name of distribution and its parameters and set up your probability no need to evaluate anything.

$$\frac{7}{10} = 0.7$$

- a. [1 point] What is the probability that exactly 7 customers arrive in a given hour?
 b. [1 point] What is the probability that the time between two consecutive customer arrivals is more than 10 minutes?
 c. [1 point] Suppose the shop is open for 30 hours, and find the probability that there are exactly 2 hours with no customers at all.

a) $X \sim \text{Pois}(\lambda=10), P(X=7)$

b) $X \sim \text{Exp}(\lambda=10), P(X > 10)$

c) $X \sim \text{Bin}(n=30, p), P(X=0)$

Problem 9. For part (a) and (b) determine whether it is true or false

- a. [0.5 point] $\text{Var}(2X-Y) = 4\text{Var}(X) + \text{Var}(Y)$. This means X and Y are independent. *False*
 b. [0.5 point] $Y = 2X$, then $\text{Cov}(Y, X)$ must be positive. *True*
 c. [0.5 point] We know as X is decreasing Y will increase, what is sign of $\text{Cov}(X, Y)$.

Negative

